SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

40.016 The Analytics Edge
Data Competition 2026

Predicting Choice Among Car Safety Bundles

Final Report · Team 3 R-izzlers

1.185	**1.185**
public leaderboard, 3rd place	private leaderboard, 4th place

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1 Introduction

Each of 1,135 survey respondents faced 19 choice tasks. A task shows three configured safety bundles and a fourth option, "none of these", and the respondent picks one. We predict the four choice probabilities for every task of 263 *new* respondents (4,997 tasks), scored by mean multiclass log loss.

Three characteristics of the data shaped our modelling decisions. First, Ch4 is the most frequently chosen alternative, accounting for 30.2% of training choices. Second, the 19 product attributes are ordinal tiers, so treating their levels as numerical imposes a linearity the data does not support. Third, the test population differs materially from training: median income rises from $60k to $80k (mean $75k to $151k), while luxury vehicles account for 69% of test tasks versus 9% in training. Transfer to unseen respondents was therefore an important part of the problem.

Our final submission combines two modelling tracks using an equal-weight geometric pool. It achieved a log-loss of 1.185 on both the public and private leaderboards, ranking 3rd and 4th respectively.

2 Validation strategy

Grouped folds

All cross-validation was grouped by respondent. Track A and the main model-comparison experiments used one fixed five-fold partition, while Track B used a respondent-grouped partition. Because each respondent answers 19 tasks, row-wise splitting would allow information about the same person to appear in both training and validation. We therefore held out complete respondents, matching the cold-start structure of the test set.

A model's score on these respondents is its out-of-fold (OOF) log loss. Blend weights are refit five times, each time excluding the fold being scored; we refer to scores produced under this procedure as *nested*.

Estimating model variation

We measured score variation across random seeds before interpreting small gains. Improvements below this variation were treated cautiously in later model comparisons.

comparison	yardstick	measured
two single models	seed sd of one model	0.00283
two blends	seed sd of the blend	0.00048
one model across folds	fold-to-fold sd	0.013

Model comparisons also use paired tests with respondent-clustered standard errors so that differences are assessed on identical held-out rows.

3 Final model

The final submission combines two modelling tracks using an equal-weight geometric pool. Track A combines listwise XGBoost with a latent-class choice model, while Track B combines latent taste features with tree and neural utility models. Figure 1 summarises the prediction pipeline.

3.1 Track A: XGBoost and latent-class choice model

Track A was built around three modelling decisions.

Part-worth coding. Allowing each attribute level to have its own utility improved OOF log-loss by approximately 0.020 ($z = 11.8$), the largest individual modelling gain we observed. The fitted price response falls sharply at first and then flattens (Figure 2), which linear coding cannot represent.

Listwise utility modelling. Representing each task as four competing alternatives improved OOF log-loss by approximately 0.006 ($z = 4.3$). XGBoost was therefore trained using a custom choice-set objective based on the within-task softmax.

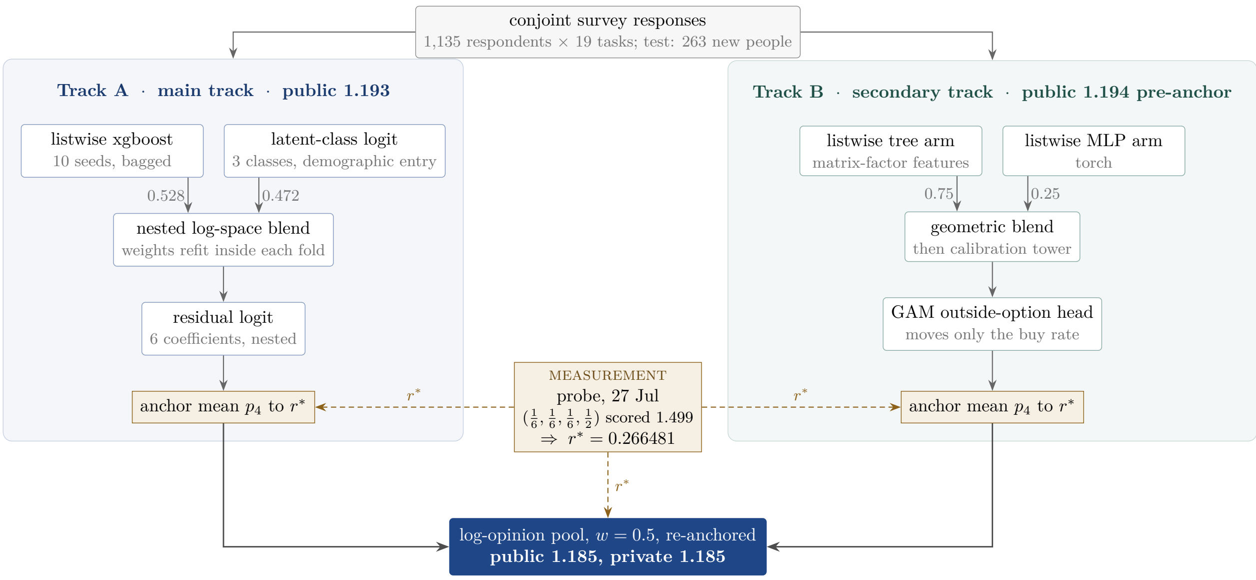

Figure 1: The graded submission, data to file. Rounded boxes are fitted models; square amber boxes are the one measured constant and the places it is applied; numbers on edges are blend weights. The pool step fits nothing: $w = 0.5$ was fixed before upload and never tuned. Track A as drawn scored public 1.193; track B's public 1.194 predates its anchor step, which was added at pooling (Section 3.3).

Latent preference heterogeneity. A three-class latent-class conditional logit improved OOF log-loss by approximately 0.013 ($z = 3.8$). Class membership is predicted from demographics only, allowing heterogeneity to be projected to unseen respondents.

The two members, a ten-seed listwise XGBoost model and the latent-class model with task-position terms, are pooled in log space with weights 0.528 and 0.472. Their original nested estimate was 1.12819; a later encoding audit showed that this was inflated by roughly 0.008, giving a leak-free nested figure close to 1.136 (Section 4).

A six-coefficient residual logit was then applied to systematic probability errors using relative price, an outside-option constant, bundle richness relative to the best rival, and interactions with luxury segment. This was the only late-stage residual correction that improved both the production folds and an independent fold structure: +0.00300 on the former and +0.00628 on the latter. The combined evidence was still modest, so we treated the correction as a small refinement rather than a major source of performance.

3.2 Track B: taste features with tree and neural models

Track B was developed as a second route to the four-choice probability distribution.

Repeated choices were first converted into respondent-level preference profiles. For each task, the selected alternative was compared with those actually offered, so the profile captured relative responses to price, safety attributes and the outside option. Aggregating these signals over 19 tasks produced a respondent-by-preference matrix.

A rank-3 matrix factorisation compressed this matrix into three latent taste coordinates. Because test respondents have no observed choice history, these coordinates could not be computed directly at prediction time. We therefore fitted demographic mappings from the observed respondent variables to the three latent coordinates and used them to project unseen respondents into the same taste space. The projected taste variables were then converted into alternative-level compatibility features for the predictive models.

The matrix-factorisation stage does not itself produce the final choice probabilities. Instead, Track B uses two four-choice utility learners: a listwise XGBoost model and a Torch multilayer perceptron. Their predictions are combined geometrically, with approximately 75% weight on XGBoost and 25% on the neural model.

A small probability-calibration stage follows. Segment-level and low-dimensional residual corrections are applied first, after which a binomial GAM supplies an additional estimate of Ch4 propensity. The GAM estimate is mixed into the existing Ch4 probability at 25%. The remaining probability mass is redistributed proportionally across Ch1–Ch3, so the relative ranking of the three purchase bundles from the primary tree–neural model is preserved.

Track B scored 1.194 publicly before the final Ch4 calibration. Its blended arms score 1.153 OOF on its own respondent-grouped partition. Its value in the final ensemble was therefore complementarity rather than superior standalone performance.

3.3 Aggregate Ch4 calibration

On 27 July, we used one submission as a diagnostic rather than as a candidate model. The file `probe_alt4.csv` predicts $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{2})$ on every row. For a constant prediction,

$$L = (1-r)\log 6 + r\log 2 \qquad \Longrightarrow \qquad r^* = \frac{\log 6 - 1.499}{\log 3} = 0.266481.$$

The returned public score therefore implied a Ch4 share of approximately 0.2665 on the public leaderboard subset. We treated this as a competition-stage calibration target rather than as a parameter estimated from training data.

Both tracks were adjusted so that their mean predicted Ch4 probability matched r^*, while preserving the relative probabilities of Ch1–Ch3. This adjustment had little effect on Track A but materially improved Track B, whose mean Ch4 probability had previously been 0.211.

3.4 Final two-track ensemble

The two tracks had similar standalone public performance but were produced through different modelling pipelines. Rather than introduce another late-stage tuned parameter, we fixed the final ensemble weight at $w = 0.5$.

Geometric pooling also provides a useful safety property. If the pooled probabilities are formed as $P_k \propto A_k^{1-w} B_k^{w}$, Hölder's inequality implies that, on a fixed evaluation set,

$$\text{loss(pool)} \le (1-w)L_A + wL_B \le \max(L_A, L_B).$$

This supported combining the tracks without a finely tuned ensemble weight.

The tracks were also meaningfully different in practice. Their within-purchase probabilities differed by 0.0129 on average, compared with a 0.0055 difference obtained from simply reseeding one model, and their predicted probability vectors differed on 96.6% of rows. This supported treating the two tracks as complementary ensemble members rather than minor variants of the same learner.

3.5 Alternative models considered

Before Track A's two blend members were chosen, nine model families were fitted under the same respondent-grouped validation framework. This comparison was useful because it separated improvements from additional model capacity from improvements that better matched the structure of the choice problem.

model family	OOF	verdict
listwise xgboost, 10-seed bag	1.13682	**blend member 1**
latent-class logit, 3 classes	1.13863	**blend member 2**
nested logit, {bundles} vs {none}	1.15681	no gain over plain logit
part-worth conditional logit	1.15686	one-class special case
elastic-net logit	1.16718	shrinkage cannot replace part-worths
two-stage buy / bundle model	1.17169	splits one utility scale in two
mixed logit	1.17281	predicts a population average
wide four-class xgboost	1.17456	cannot compare within a choice set
hierarchical Bayes	1.23703	cold-start limitation
nested blend	1.12819	honest ≈ 1.136 (Section 4)

The comparison suggests that the strongest gains came from modelling the choice-set structure directly, allowing nonlinear part-worths, and representing heterogeneity in a way that could still be estimated for unseen respondents. More flexible models were not automatically better under the cold-start evaluation.

4 Model development and rejected experiments

Several rejected experiments materially changed how we evaluated later modelling decisions.

4.1 Seed variation and apparent gains

Refitting the same XGBoost configuration under ten seeds moved OOF log-loss across a range of 0.0090 (sd 0.00283). This invalidated one apparent gain from a monotone price constraint: an initial improvement of $+0.00172$ became -0.00034 under a paired ten-seed comparison, with 95% CI $[-0.00159, +0.00092]$. The constrained variant was therefore removed.

Averaging predetermined seeds reduced Monte-Carlo variation without introducing another validation-selected hyperparameter.

4.2 Leakage despite respondent-grouped folds

One derived residual feature produced an unusually large apparent improvement of about 0.03. Although respondent folds were separated correctly, an upstream baseline prediction had indirectly used information from the scored fold. Rebuilding the feature while excluding both the reference and evaluation folds removed the gain (-0.00430, $z = -4.54$).

A second issue affected a design-share encoding constructed before the CV loop. Because the encoding was generated once and then reused inside cross-validation, training rows could contain summaries influenced by the scored fold even though the respondent split itself was correct. The feature initially appeared to improve OOF loss by about 0.022 and remained in the production pipeline for many iterations.

Three checks eventually exposed the problem. First, an honestly reconstructed version produced a negative value of -0.00596 ($z = -3.53$). Second, the apparent gain increased with tree depth, from approximately -0.0004 at depth 4 to $+0.078$ at depth 10, which is more consistent with a flexible model exploiting leakage than with a stable predictive effect. Third, removing the feature changed the ranking of the earlier hyperparameter sweep. These cases showed that grouped folds alone are insufficient if feature construction itself uses information from the evaluation fold.

4.3 How this changed our validation process

After the leakage audit, small improvements in the original OOF metric were treated cautiously. A segment-reweighted OOF score was used as a secondary diagnostic because it tracked the public leaderboard more closely when leak-free, but its effective sample of 208 respondents made it unsuitable as a training objective.

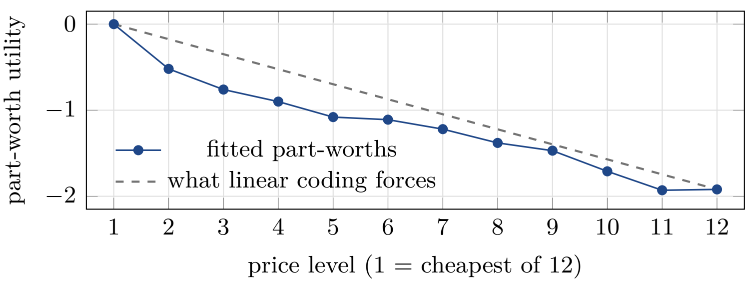

Figure 2: Price response flattens: the first step up from the cheapest level costs 0.52 utility, later steps average 0.13, and the top two levels are nearly identical. Freeing this shape was worth +0.020 log loss.

5 Model insights

5.1 Latent preference groups and price response

The latent-class model splits respondents into three near-equal segments. Membership is predicted from demographics, with annual mileage, night driving, education and parking situation among the strongest drivers:

segment	share	price response	outside-option constant
price-sensitive group	36.6%	−1.65, steady decline	−1.40
weakly price-sensitive / premium-oriented group	32.1%	+0.50, flat to positive	−1.30
opt-out-prone group	31.4%	−0.60, mild	+0.17

The aggregate price curve flattens at higher price levels, but the latent classes suggest that part of this shape comes from mixing respondents with different price sensitivities. The first group shows a steep and consistently negative price response, while the second is close to flat over much of the range and the third is comparatively mild. Averaging these curves naturally produces a flatter aggregate response even when no single segment has exactly that shape.

The positive price response in the second class may reflect premium orientation or residual confounding between price and bundle richness, so it should not be interpreted causally. The third class is notable for its higher baseline tendency to choose Ch4. Commercially, this suggests that price changes alone are unlikely to affect every respondent group equally.

5.2 Discrete heterogeneity transferred better under cold start

Mixed logit estimated substantial heterogeneity but scored 1.173, while hierarchical Bayes scored 1.23703 overall. The cold-start problem is visible when the same HB model is scored using respondent-specific fitted tastes versus population-average tastes:

prediction for the same rows	log loss
each respondent's own fitted tastes	0.352
population-averaged tastes	1.229

The 0.877 difference shows how much respondent-specific information becomes unavailable for unseen test respondents. Hierarchical Bayes is powerful when a respondent's own choice history is available, but much of that advantage disappears when predicting for a new person.

The latent-class model transferred better because it routes heterogeneity through a much lower-dimensional structure that can be estimated from demographics available at prediction time. Removing demographics improved HB from 1.237 to 1.164, while the same information accounted for 93% of the latent-class gain. This does not mean continuous heterogeneity is inherently unsuitable; here, the discrete representation provided a better bias–variance trade-off under cold start.

5.3 Separating opt-out did not improve predictive fit

A two-stage model that first predicted buy/no-buy and then bundle choice scored 1.172, and a nested-logit treatment of the outside option matched plain multinomial logit to three decimals. For this dataset, the unified utility specification was therefore sufficient.

5.4 Task position and changing price sensitivity

The walk-away rate rises across the 19 tasks of a survey session (Figure 3). Adding task-position terms to the latent-class model improved OOF log-loss by +0.0053 ($z = 6.3$) and replicated on an independent fold structure (+0.0042, $z = 4.7$). We also compared this explanation with a pure scale-drift specification in which later answers simply become noisier. The changing-price-sensitivity specification reproduced essentially all of the observed increase in Ch4, whereas scale drift reproduced very little. The fitted interaction therefore suggests that later responses are associated with stronger price sensitivity rather than simply greater response noise.

5.5 Population shift and transfer to the test set

The latent-class model predicted a lower Ch4 rate for the wealthier test population (0.223 versus 0.302 in training). The public diagnostic implied 0.266, confirming the direction but showing that the model overstated the magnitude. We also checked whether the main modelling gains survived reweighting toward the test income distribution: part-worth coding retained about 101% of its value, the listwise objective 112%, and the latent-class model 79%. This suggested that the first two improvements transferred well, while the heterogeneity model was more sensitive to population composition. The gap between the model-implied Ch4 rate and the public diagnostic motivated the final aggregate Ch4 calibration.

5.6 Remaining sources of prediction error

An oracle using each respondent's own observed choice rates scores 0.986 against 1.130 for the blend as it stood when measured, leaving a substantial gap associated with person-specific behaviour. A variance decomposition shows that Ch4 is much more personal than the three bundle choices: roughly 33% of its variation is between respondents, compared with about 5–7% for the bundles.

Demographics explain only 11.2% of that between-respondent variation in Ch4. In other words, most of the stable individual tendency to opt out is not recoverable from the variables supplied in the competition. This places a practical limit on the amount of respondent-specific variation that any model using the available features can recover.

6 Public and private leaderboard performance

6.1 Local validation versus leaderboard performance

local nested OOF	public	offset	transfer of local gain
1.17683	1.223	+0.046	first submission
1.13878	1.201	+0.062	∼58%
1.13556	1.201	+0.065	not visible
1.13044	1.199	+0.069	∼24–39%

Local OOF scores were consistently lower than public leaderboard loss by approximately 0.05–0.07, reflecting both population differences and accumulated model-selection optimism. Transfer of small local gains weakened over time, so the leaderboard was used mainly for limited aggregate diagnostics rather than repeated fine-grained model selection.

6.2 Pre-upload score forecasts

We recorded score forecasts before the corresponding submissions were evaluated, giving a prospective check on the validation and calibration logic.

quantity	forecast	band	returned
Track A final file	1.1930	point forecast	1.193
graded pool	1.186	1.184–1.189	**1.185**

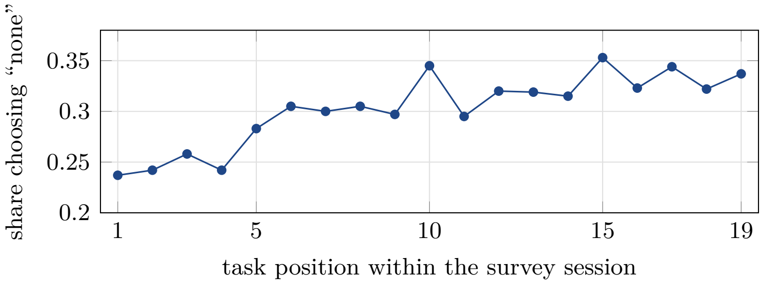

Figure 3: Raw training data: the share choosing "none" rises from 23.7% on the first task to above 33% late in the session.

6.3 Private leaderboard results

The final private score was also 1.185, showing no detectable deterioration at the leaderboard's displayed precision. This is consistent with the conservative final ensemble and with the fact that the aggregate Ch4 calibration, estimated only from the public subset, did not produce an observable loss increase on the private rows.

The rank moved from 3rd to 4th despite the unchanged score. Because all teams are scored on the same private rows, rank differences depend on where models disagree rather than on the absolute sampling variability of one team's score alone. We therefore interpret the unchanged loss as evidence of stable transfer, while treating the one-place rank change as compatible with ordinary paired ranking variation.

7 Limitations

A substantial amount of respondent-level variation remains unexplained by the available variables. Some gains may not transfer directly to a different experimental design or population, particularly those routed through choice-set identity and latent-class membership. Evidence for preference heterogeneity was consistent, but the exact class boundaries were less stable: several nearby class specifications fit similarly, so the three segments should be treated as a useful predictive summary rather than as uniquely identified consumer types.

The submitted tree pipeline still contains the design-share feature discussed in Section 4. Subsequent ablation suggested that removing it was approximately leaderboard-neutral, but we did not rebuild the final candidate late in the submission window. We would exclude this feature in a future implementation.

Finally, we optimise average log loss only; no subgroup is separately calibrated, and predictions are least reliable where support is thinnest, particularly the wealthy luxury segments concentrated in the test set.

8 Reproducibility

Fresh training approximately reconstructs both modelling procedures. Track A can be rerun from its modelling script followed by the residual-logit and calibration steps, while Track B has its own end-to-end runner. Multithreaded XGBoost and Torch may produce small numerical differences across machines, so fresh retraining is not expected to be bit-identical.

Separately, a deterministic reconstruction using `build_candidates.R`, the pinned Track A and Track B outputs, and the Ch4 calibration agrees with the scored submission to within 1.1×10^{-15}. Two implementation details are documented because they affected reproduction: XGBoost 3.x returns a matrix from `predict` where older versions returned a vector, and conditional-logit designs must place reference levels carefully because variables constant within a choice set are unidentified.

9 Conclusion

Three modelling choices made early in the project—part-worth coding, a listwise objective, and demographically routed discrete heterogeneity—accounted for most of the predictive performance we achieved. Later work contributed mainly by clarifying where the remaining error lies, identifying two sources of leakage, and improving how we interpreted small validation gains.

The final submission used a fixed two-track ensemble and achieved a log-loss of 1.185 on both the public and private subsets. Three lessons were most useful in retrospect: estimate stochastic variation before acting on small gains, construct leak-free comparisons for features that appear unusually strong, and validate on respondents excluded from fitting when the test set consists of unseen respondents.